MATH1560J Final Exam Practice Set

Problem 1. Given the parametric curve $(x(t), y(t))$ where:

$$x(t) = 2\cos(3t) - 3\sin(2t), \quad y(t) = 4\sin(3t) + 3\cos(4t)$$

Find the equations (in terms of x, y only) for tangent lines at $t = 0$ and $t = \pi$.

Problem 2. Let $R \subseteq \mathbb{R}^2$ be the region bounded by the graphs of functions $y = f(x) = x^2$ and $y = g(x) = 4 - 2x - x^2$.

(i) Write down the integral for calculating the **perimeter** (total length of boundary) of R. Then evaluate the integral.

(ii) Let E be the solid obtained by rotating the region R around the x-axis. Write down the integral for calculating the **surface area** of E. Then evaluate the integral.

(Remark: Briefly explain your formula if you consider it too exotic for others to understand.)

Problem 3. Given the function:

$$f(x) = \sum_{k\geq 0} \left(\frac{3}{5^k} + (-1)^k \frac{5}{3^k} \right) x^k$$

Find the Taylor series for $f(x)$ centered at $x = 1$. Show your work.

Problem 4. Given $f(x) = \frac{x}{\log(1-x)}$ for $0 < x < 1$.

Evaluate the following series for $x \in (0, 1)$:

$$\sum_{n \geq 1} \frac{x^n (1-x)^n}{n!} f^{(n)}(x)$$

(Your result should be a formula in terms of x and does not contain infinite summation.)

Problem 5. Find the area that is both **inside** the polar curve $r = 1 + \cos\theta$ (cardioid) and **outside** the polar curve $r = \sqrt{3}\sin\theta$ (circle).

Note: Write down the integral for calculating the area first, then evaluate the integral.
(Remark: Briefly explain your formula if you consider it too exotic for others to understand.)

Problem 6. Given the power series:

$$\sum_{n\geq 1} \frac{2^n (x-10)^n}{n^{1/3}}$$

(i) Find the radius of convergence (ROC) for the series. Show your work and identify convergence tests you applied.

(ii) Find the interval of convergence (IOC) for the series. Show your work and identify convergence tests you applied.

Problem 7. Evaluate the following definite integral:

$$\int_0^2 \arcsin\left(\frac{2x}{1+x^2}\right)\, dx$$

Problem 8. Find the antiderivative of f, where:

$$f(x) = \frac{x^4 + 4x^3 + 11x^2 + 12x + 8}{(x^2 + 2x + 3)^2(x + 1)}$$

Appendix: Useful Power Series Expansions

- $\dfrac{1}{1-x} = \sum_{n\geq 0} x^n, \quad \text{ROC} = 1$
- $e^x = \sum_{n\geq 0} \dfrac{x^n}{n!}, \quad \text{ROC} = \infty$
- $\sin x = \sum_{n\geq 0} (-1)^n \dfrac{x^{2n+1}}{(2n+1)!}, \quad \text{ROC} = \infty$
- $\cos x = \sum_{n\geq 0} (-1)^n \dfrac{x^{2n}}{(2n)!}, \quad \text{ROC} = \infty$
- $\arctan x = \sum_{n\geq 0} (-1)^n \dfrac{x^{2n+1}}{2n+1}, \quad \text{ROC} = 1$
- $\log(1+x) = \sum_{n\geq 1} (-1)^{n-1} \dfrac{x^n}{n}, \quad \text{ROC} = 1$
- $(1+x)^\alpha = \sum_{n\geq 0} \binom{\alpha}{n} x^n, \quad \text{ROC} = 1$